Genome-level antibiotic-resistance prediction

E. coli, 37 antibiotics - ESM-2 protein embeddings with Transformer, CNN and BiLSTM heads

Headline

Reading every protein of a genome through a pretrained protein language model (ESM-2 35M) lifts resistance prediction from near chance to strong. The earlier model that saw 64 random proteins per genome reached macro AUROC 0.59. Heads that read all ~4,700 proteins reach 0.90-0.93.

Data

- BacBench antibiotic-resistance protein sequences (Hugging Face), filtered to E. coli (taxid 562): 7,211 genomes, of which 6,055 carry at least one resistance label.
- Labels: resistant / susceptible for 37 antibiotics (binary_labels.csv). A genome is usually tested against only some antibiotics; missing labels are ignored in the loss.
- Split by genome, random, seed 42: 5,450 train / 605 validation.
- 29.0 million protein occurrences, of which only 1.35 million are unique sequences (4.7%). Each unique protein is embedded once.

Method

1. Embed: each unique protein goes through frozen ESM-2 35M (fp16); residue vectors are mean-pooled to 480 numbers. About 32 min on one A10 GPU.
2. Read the genome: a trained head takes all ~4,700 protein vectors of a genome and outputs 37 resistance probabilities.
3. Train: masked binary cross-entropy with per-antibiotic positive weighting; AdamW; best epoch picked by validation AUPRC.
4. Score: AUROC and AUPRC per antibiotic; headline = macro average over the 21 antibiotics with at least 10 resistant validation genomes.

The five heads

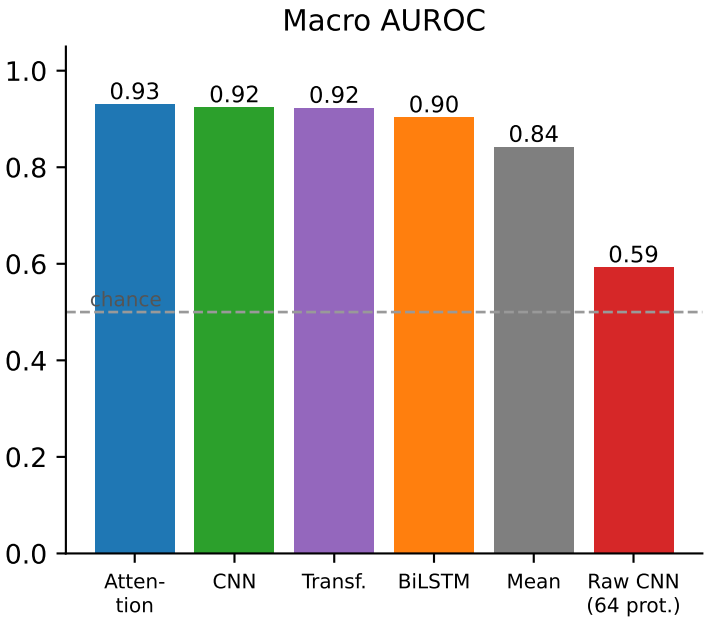
- Attention: learns a relevance weight per protein (separately for each antibiotic) and takes a weighted sum. Proteins are treated as a set.
- Transformer: two self-attention layers let proteins look at each other, then mean+max pooling. Set input; trained on random 1,024-protein subsets.
- CNN: 1D convolutions (kernels 3/5/7/9) and max pooling along the genome, proteins in genome order.
- BiLSTM: bidirectional LSTM along the genome, proteins in genome order.
- Mean pooling: average all protein vectors, then a linear layer (baseline).

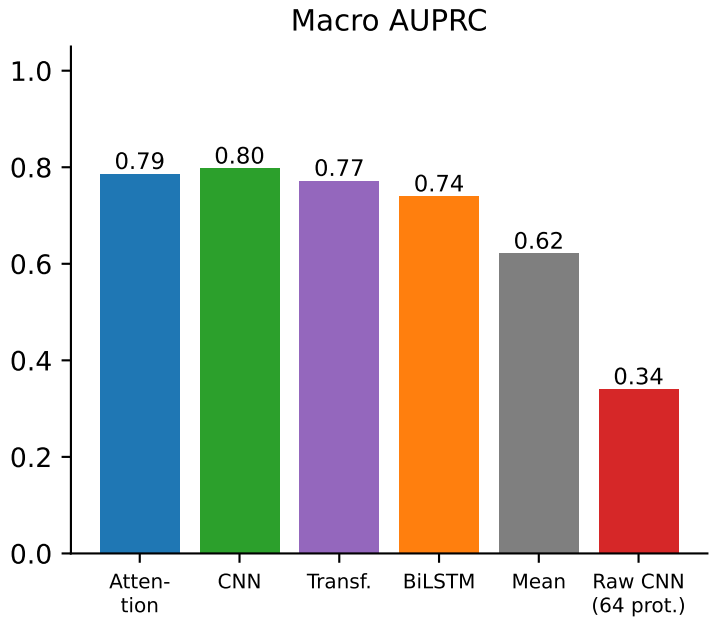
Metrics in one line each

AUROC: chance that a random resistant genome scores higher than a random susceptible one (0.5 = coin flip, 1.0 = perfect).
AUPRC: area under the precision-recall curve; its chance level equals the share of resistant genomes, so it is stricter when resistance is rare.

Macro results (21 antibiotics, validation genomes)

Model	AUROC	AUPRC	Best epoch
Attention	0.931	0.786	30
CNN	0.925	0.798	23
Transformer	0.923	0.772	23
BiLSTM (+10 ep.)	0.912	0.753	30+10
BiLSTM (30 ep.)	0.904	0.740	30
Mean pooling	0.843	0.621	30
Raw-seq CNN, 64 prot.	0.593	0.340	8



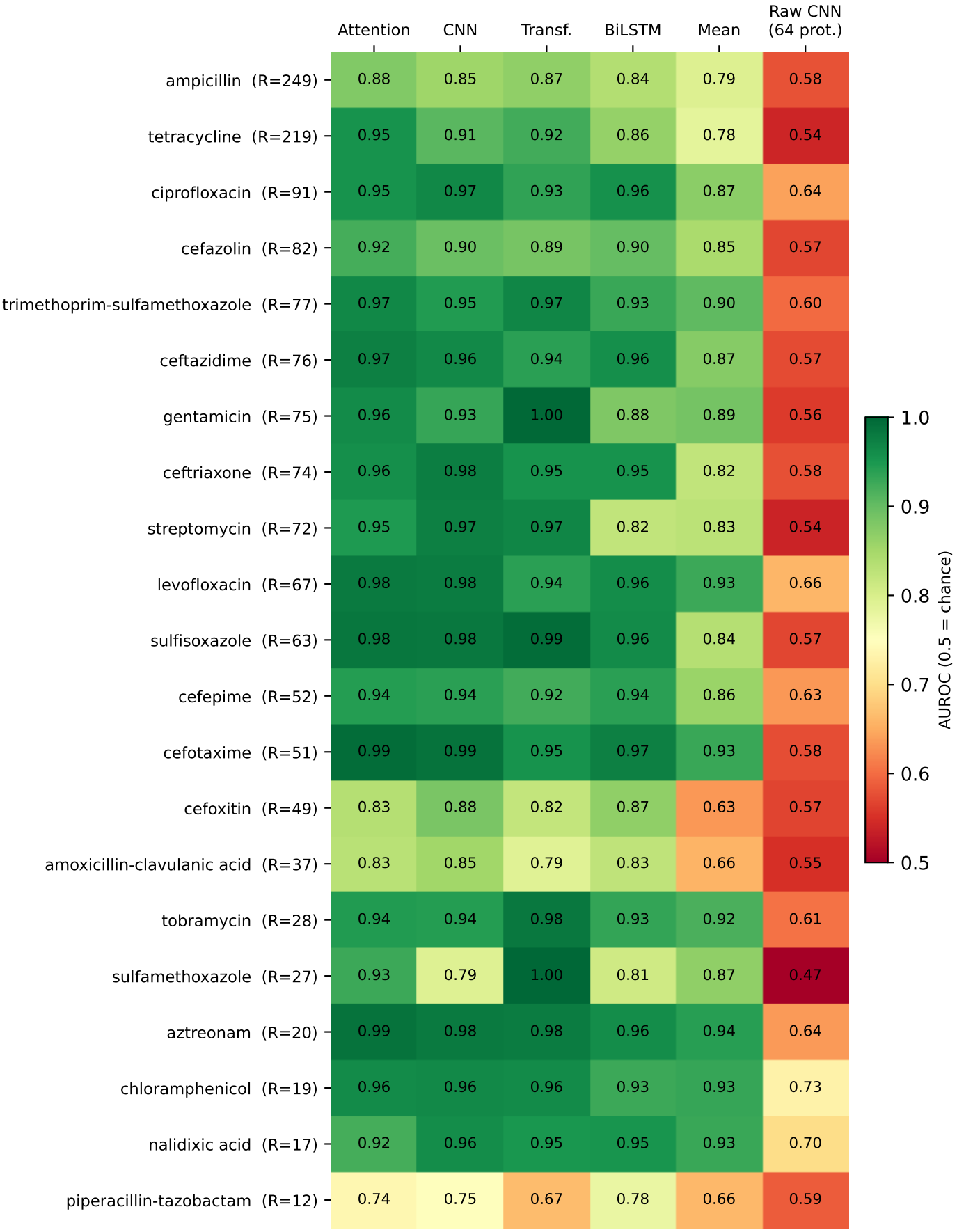


What the figure shows

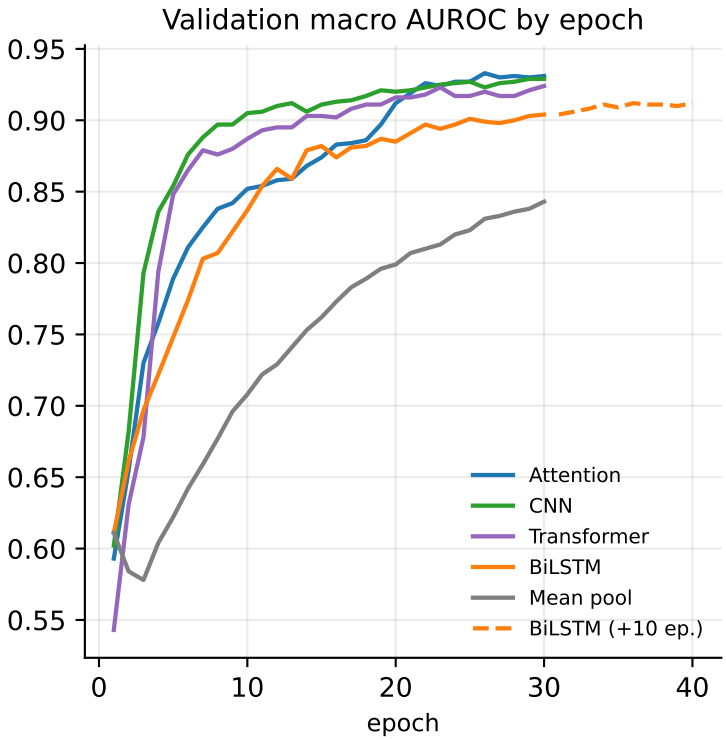
- The jump from the raw-sequence CNN (0.59) to every ESM-based head (0.84-0.93) is far larger than any noise could explain. Seeing all proteins matters most.
- Attention, CNN and Transformer are within 0.01 AUROC of each other; with one run each that gap is within noise, so no winner among the three can be claimed.
- Mean pooling is clearly worse (0.84): averaging ~4,700 proteins dilutes the few that carry resistance.
- BiLSTM gained +0.008 AUROC / +0.013 AUPRC from 10 more epochs and still trails the other three, mainly in AUPRC.
- The raw-sequence CNN used a random 64 of ~4,500 proteins per genome and cannot see most resistance genes.

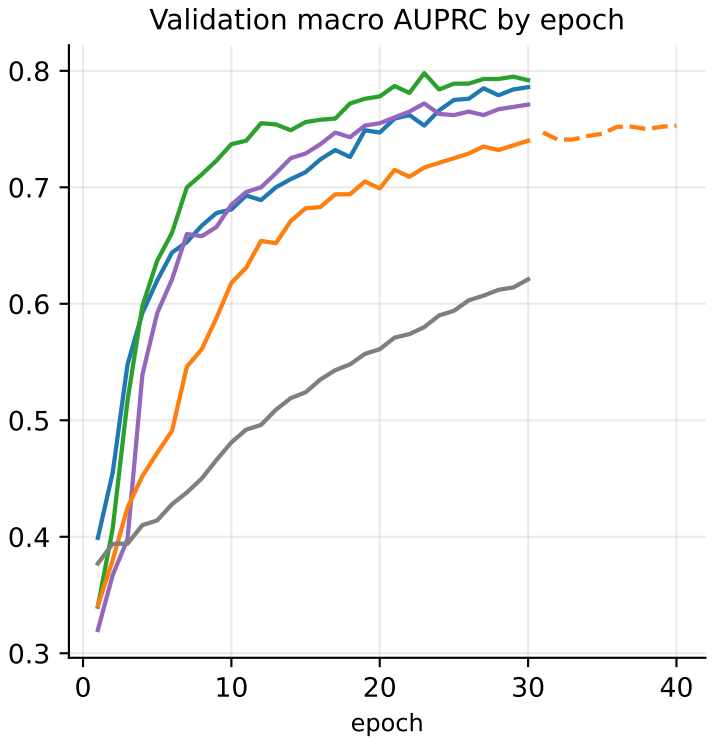
Per-antibiotic AUROC

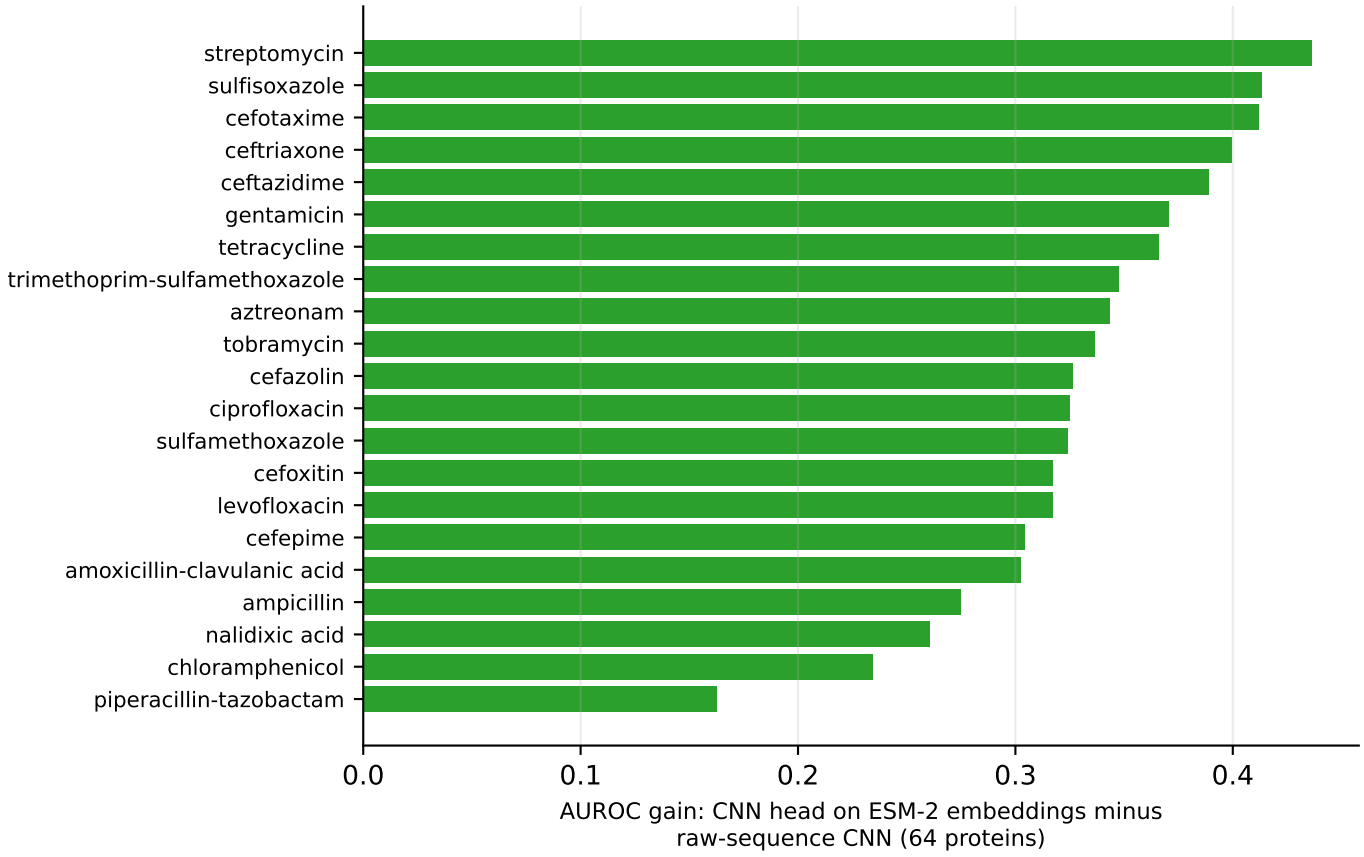
Antibiotics with at least 10 resistant validation genomes, sorted by number of resistant genomes (R).



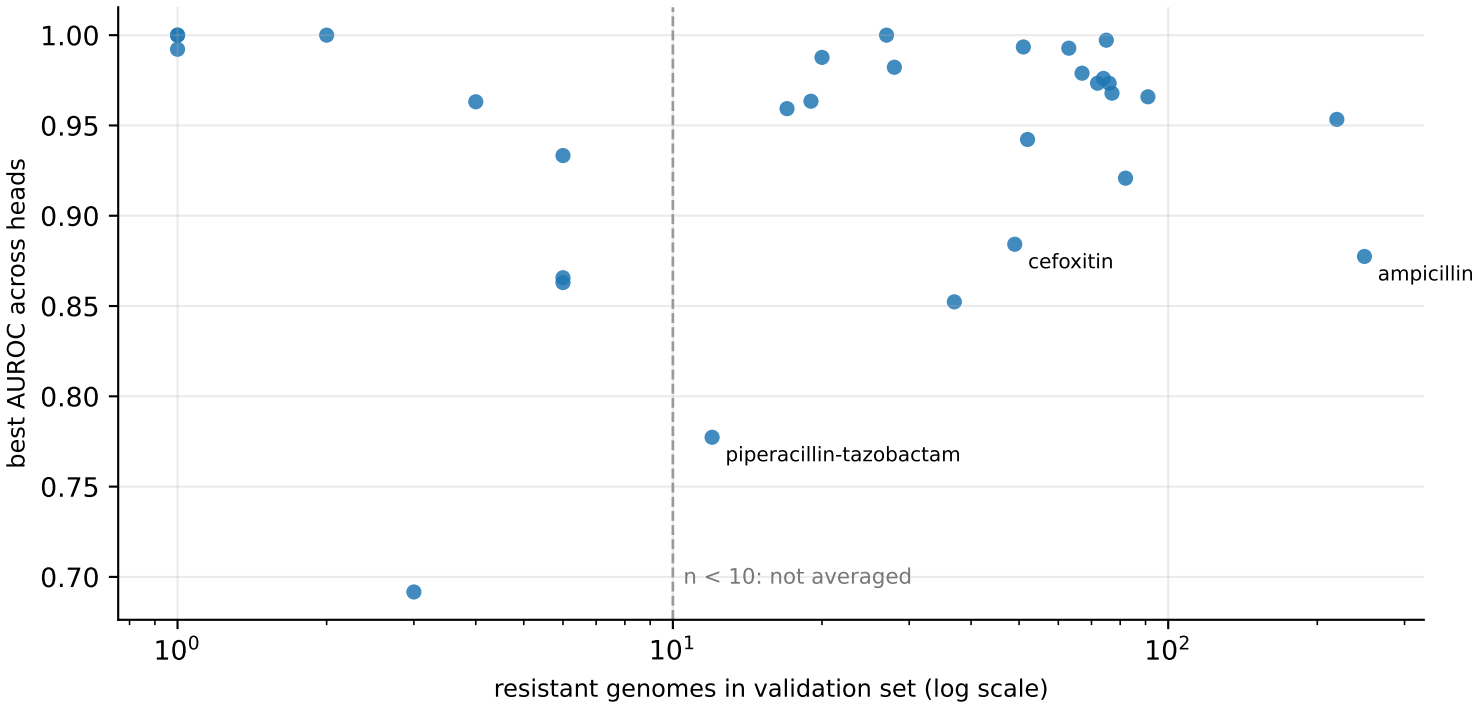
Training curves and gain over the raw-sequence model







How much data each antibiotic has, and what to trust



Caveats - please read before citing these numbers

- Single runs, one seed each. Differences of about 0.01 between heads are within noise.
- No held-out test set. The best epoch was chosen on the same 605 validation genomes the scores are reported on, so they are slightly optimistic.
- Random split, not lineage-aware. Near-identical strains (e.g. from one outbreak) can sit in both train and validation and inflate scores.
- BiLSTM, attention and mean heads were still improving at the 30-epoch cap.
- No simple baseline (such as gene presence/absence with logistic regression) has been run, so we cannot yet say how much of the result needs a neural network.
- Weak spots: ampicillin (0.84-0.88), cefoxitin, amoxicillin-clavulanic acid, and piperacillin-tazobactam (only 12 resistant genomes).

Next steps (deferred)

1. Held-out test split, ideally lineage-aware.
2. Gene presence/absence baseline.
3. Several seeds per head, longer training with early stopping.
4. Optional: larger ESM-2 models (150M / 650M).

Reproducibility

Code: embed.py, order.py, amr_embed.py (heads), amr.py (raw-sequence baseline). Trained heads, metrics, logs and split are in results/. Embeddings (1.3 GB) regenerate with embed.py in about 30 minutes on one A10. Run cost: about $1 of GPU time.